

Vector Decomposition

Tanmay Kapure

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1 Question 1

Consider the vectors

$$u = \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} \quad \text{and} \quad v = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}.$$

(a) Decomposition of u into two orthogonal vectors

1. Projection of u onto v :

The projection of u onto v is given by:

$$\text{proj}_v(u) = \frac{u \cdot v}{v \cdot v} v.$$

First, we compute the dot products:

$$u \cdot v = 2 \cdot 1 + 4 \cdot 0 + 0 \cdot 3 = 2,$$

$$v \cdot v = 1^2 + 0^2 + 3^2 = 1 + 9 = 10.$$

Now, the projection of u onto v is:

$$\text{proj}_v(u) = \frac{2}{10} v = \frac{1}{5} \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} \frac{1}{5} \\ 0 \\ \frac{3}{5} \end{pmatrix}.$$

2. Orthogonal component of u to v :

The orthogonal component is found by subtracting the projection from u :

$$u_{\perp} = u - \text{proj}_v(u) = \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} - \begin{pmatrix} \frac{1}{5} \\ 0 \\ \frac{3}{5} \end{pmatrix}$$
$$u_{\perp} = \begin{pmatrix} 2 - \frac{1}{5} \\ 4 - 0 \\ 0 - \frac{3}{5} \end{pmatrix} = \begin{pmatrix} \frac{9}{5} \\ 4 \\ -\frac{3}{5} \end{pmatrix}.$$

Thus, u is decomposed as:

$$u = \text{proj}_v(u) + u_{\perp} = \begin{pmatrix} \frac{1}{5} \\ 0 \\ \frac{3}{5} \end{pmatrix} + \begin{pmatrix} \frac{9}{5} \\ 4 \\ -\frac{3}{5} \end{pmatrix}.$$

(b) Decomposition of v into two orthogonal vectors

1. Projection of v onto u :

The projection of v onto u is:

$$\text{proj}_u(v) = \frac{v \cdot u}{u \cdot u} u.$$

First, we compute the dot products:

$$v \cdot u = 1 \cdot 2 + 0 \cdot 4 + 3 \cdot 0 = 2,$$

$$u \cdot u = 2^2 + 4^2 + 0^2 = 4 + 16 = 20.$$

Now, the projection of v onto u is:

$$\text{proj}_u(v) = \frac{2}{20} u = \frac{1}{10} \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{5} \\ \frac{2}{5} \\ 0 \end{pmatrix}.$$

2. Orthogonal component of v to u :

The orthogonal component is:

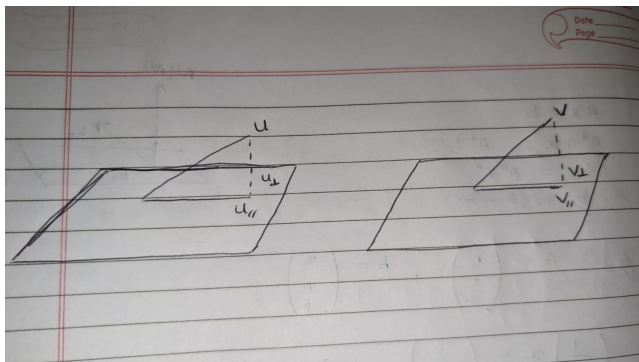
$$v_{\perp} = v - \text{proj}_u(v) = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} - \begin{pmatrix} \frac{1}{5} \\ \frac{2}{5} \\ 0 \end{pmatrix}$$

$$v_{\perp} = \begin{pmatrix} 1 - \frac{1}{5} \\ 0 - \frac{2}{5} \\ 3 - 0 \end{pmatrix} = \begin{pmatrix} \frac{4}{5} \\ -\frac{2}{5} \\ 3 \end{pmatrix}.$$

Thus, v is decomposed as:

$$v = \text{proj}_u(v) + v_{\perp} = \begin{pmatrix} \frac{1}{5} \\ \frac{2}{5} \\ 0 \end{pmatrix} + \begin{pmatrix} \frac{4}{5} \\ -\frac{2}{5} \\ 3 \end{pmatrix}.$$

(c) Sketch



2 Question 2

Consider the vectors

$$u = \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \quad \text{and} \quad w = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix}.$$

(a) Decompose the vector $w = w_1 + w_2$ such that $w_1 \in \text{span}\{u, v\}$ and w_2 has minimum length.

We need to express w_1 as a linear combination of u and v , i.e., $w_1 = \alpha u + \beta v$, where α and β are scalars.

To minimize w_2 , we project w onto the subspace spanned by u and v . Using the projection formula for multiple vectors:

$$w_1 = P_{\text{span}\{u,v\}}(w),$$

where the projection matrix onto the span of u and v is given by:

$$P_{\text{span}\{u,v\}} = A(A^T A)^{-1} A^T,$$

$$\text{with } A = (u \ v) = \begin{pmatrix} 2 & 1 \\ 4 & 0 \\ 0 & 3 \end{pmatrix}.$$

We first compute $A^T A$ and then $A^T w$:

$$A^T A = \begin{pmatrix} 2 & 4 & 0 \\ 1 & 0 & 3 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 4 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 20 & 2 \\ 2 & 10 \end{pmatrix}.$$

$$A^T w = \begin{pmatrix} 2 & 4 & 0 \\ 1 & 0 & 3 \end{pmatrix} \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 9 \end{pmatrix}.$$

Now, solving the linear system:

$$(A^T A)^{-1} = \frac{1}{196} \begin{pmatrix} 10 & -2 \\ -2 & 20 \end{pmatrix},$$

we get:

$$w_1 = A(A^T A)^{-1} A^T w = \begin{pmatrix} 2 & 1 \\ 4 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 6 \\ 9 \end{pmatrix}.$$

$$w_1 = \begin{pmatrix} \frac{3}{5} \\ \frac{4}{5} \\ 2 \end{pmatrix}.$$

Thus, the decomposed vector $w_1 \in \text{span}\{u, v\}$.

(b) What is the length of w_2 ?

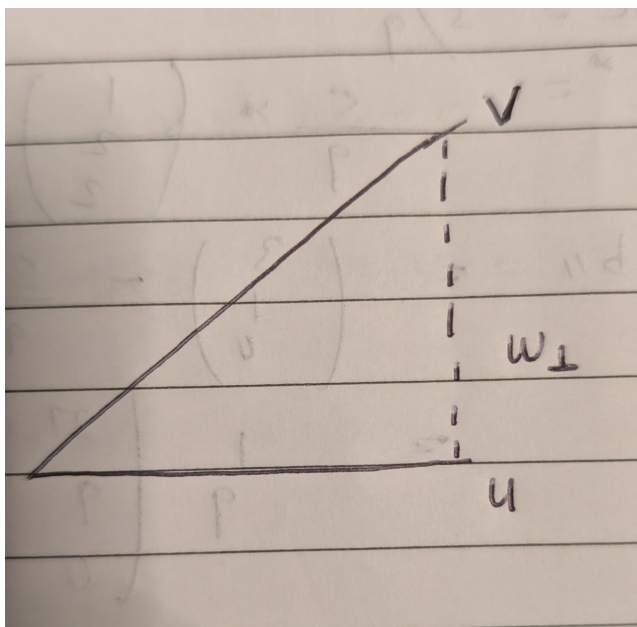
The vector w_2 is given by: $w_2 = w - w_1 = \begin{pmatrix} \frac{12}{5} \\ -\frac{4}{5} \\ 0 \end{pmatrix}$. The length of w_2 is:

$$\|w_2\| = \sqrt{\left(\frac{12}{5}\right)^2 + \left(-\frac{4}{5}\right)^2} = \sqrt{6.4} \approx 2.53.$$

Thus, the length of w_2 is approximately 2.53.

(c) Sketch of the vectors

The sketch will show the vectors u , v , and w , along with their projections in 3D space.



(d) Obtain a basis $\{e_1, e_2, e_3\}$ such that two basis vectors lie in the plane $\text{span}\{u, v\}$.

We begin by normalizing u to obtain the first basis vector:

$$e_1 = \frac{u}{\|u\|} = \frac{1}{\sqrt{2^2 + 4^2 + 0^2}} \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{20}} \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{2}{\sqrt{20}} \\ \frac{4}{\sqrt{20}} \\ 0 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}.$$

Next, we make v orthogonal to e_1 by subtracting the projection of v onto u :

$$\text{proj}_u(v) = \frac{u \cdot v}{u \cdot u} u = \frac{2(1) + 4(0) + 0(3)}{20} \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} = \frac{2}{20} \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix}.$$

Now, subtract this from v to get e_2 :

$$e_2 = v - \text{proj}_u(v) = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} - \begin{pmatrix} \frac{1}{5} \\ \frac{2}{5} \\ 0 \end{pmatrix} = \begin{pmatrix} 1 - \frac{1}{5} \\ 0 - \frac{2}{5} \\ 3 - 0 \end{pmatrix} = \begin{pmatrix} \frac{4}{5} \\ -\frac{2}{5} \\ 3 \end{pmatrix}.$$

Now normalize e_2 to get the second basis vector:

$$e_2 = \frac{1}{\sqrt{\left(\frac{4}{5}\right)^2 + \left(-\frac{2}{5}\right)^2 + 3^2}} \begin{pmatrix} \frac{4}{5} \\ -\frac{2}{5} \\ 3 \end{pmatrix} = \frac{1}{\sqrt{\frac{16}{25} + \frac{4}{25} + 9}} \begin{pmatrix} \frac{4}{5} \\ -\frac{2}{5} \\ 3 \end{pmatrix} = \frac{1}{\sqrt{10}} \begin{pmatrix} \frac{4}{5} \\ -\frac{2}{5} \\ 3 \end{pmatrix}.$$

We can find e_3 as the cross product of e_1 and e_2 :

$$e_3 = e_1 \times e_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} & 0 \\ \frac{4}{\sqrt{10}} & -\frac{2}{\sqrt{10}} & \frac{3}{\sqrt{10}} \end{vmatrix}.$$

Expanding this determinant:

$$e_3 = \mathbf{i} \left(\frac{2}{\sqrt{5}} \cdot \frac{3}{\sqrt{10}} - 0 \right) - \mathbf{j} \left(\frac{1}{\sqrt{5}} \cdot \frac{3}{\sqrt{10}} - 0 \right) + \mathbf{k} \left(\frac{1}{\sqrt{5}} \cdot \left(-\frac{2}{\sqrt{10}} \right) - \frac{2}{\sqrt{5}} \cdot \frac{4}{\sqrt{10}} \right).$$

Simplifying, we get:

$$e_3 = \begin{pmatrix} \frac{6}{\sqrt{50}} \\ -\frac{3}{\sqrt{50}} \\ -\frac{10}{\sqrt{50}} \end{pmatrix} = \frac{1}{\sqrt{50}} \begin{pmatrix} 6 \\ -3 \\ -10 \end{pmatrix}.$$

Thus, the orthogonal basis is:

$$e_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad e_2 = \frac{1}{\sqrt{10}} \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix}, \quad e_3 = \frac{1}{\sqrt{50}} \begin{pmatrix} 6 \\ -3 \\ -10 \end{pmatrix}.$$

(e) Use the normal equation to express the vector w in this basis.

Given the vector

$$w = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix},$$

and the basis vectors

$$e_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad e_2 = \frac{1}{\sqrt{10}} \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix}, \quad e_3 = \frac{1}{\sqrt{50}} \begin{pmatrix} 6 \\ -3 \\ -10 \end{pmatrix}.$$

The matrix A is formed as:

$$A = (e_1 \quad e_2 \quad e_3).$$

The normal equations are given by:

$$A^T A x = A^T w.$$

Calculating $A^T A$ yields:

$$A^T A = I_3 \quad (\text{identity matrix}).$$

Calculating $A^T w$:

$$A^T w = \begin{pmatrix} \frac{9}{5} \\ \frac{12}{5} \\ -2 \end{pmatrix}.$$

Thus, solving for coefficients gives:

$$x = A^T w = \begin{pmatrix} 1.8 \\ 2.4 \\ -2 \end{pmatrix}.$$

Therefore, we express w as:

$$w = x_1 e_1 + x_2 e_2 + x_3 e_3 = 1.8 \cdot e_1 + 2.4 \cdot e_2 - 2 \cdot e_3.$$